

**A Bayesian Coalescent Model of the DNA Barcode Gap**

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**Running Title:** Bayesian inference for DNA barcode gap coalescent estimation

## Abstract

A simple statistical model of the DNA barcode gap is outlined. Specifically, accuracy of recently introduced nonparametric metrics, inspired by coalescent theory, to characterize the extent of proportional overlap/separation in maximum and minimum pairwise genetic distances within and among species, respectively, is explored in both frequentist and Bayesian contexts. The empirical cumulative distribution function (ECDF) is utilized to estimate probabilities associated with positively skewed extreme tail distance distribution quantiles bounded on the closed unit interval  $[0, 1]$  based on a straightforward binomial count of overlapping specimen records. Statistical moments, notably, the mean and variance, are derived for edge cases of no overlap/complete separation and full overlap/no separation in genomic sequence variation, and are shown to be quite tight as sample sizes increase. Further, a new way to visualize distances and the DNA barcode gap estimators via accumulated probabilities appears revealing.

Using R and the probabilistic programming language, Stan, the proposed maximum likelihood estimators and Bayesian model are demonstrated on cytochrome *b* (CYTB) gene sequences from two *Agabus* diving beetle species, *A. bipusulatus* and *A. nevadensis*, exhibiting limits in the extent of representative taxonomic sampling ( $N = 701$  and  $N = 2$  individuals, respectively). Large-sample theory and MCMC simulations clearly expose problems, showing much uncertainty in parameter estimates, particularly under the former paradigm, and when specimen sample sizes for target species are small. Findings herein highlight the promise of the Bayesian approach using a conjugate beta prior for reliable posterior estimation over classical inference when available data are sparse. Obtained results can help shed light on foundational and applied research questions concerning DNA-based specimen identification and species delineation for studies in evolutionary biology and ecology, as well as biodiversity conservation, forensics, management and restoration, of wide-ranging taxa.

**Keywords:** Bayesian/frequentist inference, DNA barcoding, Hamiltonian Monte Carlo (HMC), intraspecific genetic distance, interspecific genetic distance, multispecies coalescent, specimen identification, species discovery

# 1 Introduction

The routine use of DNA sequences to support broad evolutionary hypotheses and questions concerning central demographic processes, like gene flow and speciation, is now ubiquitous. Since the late 1980s, mitochondrial DNA (mtDNA) has been employed to produce a distinctive and measurable signature of genetic polymorphism that readily elucidates phylogeographic patterns like colonization and extinction in diverse and spatially-distributed taxonomic lineages, such as birds, fishes, insects, and arachnids, among other extensively studied groups (Avice et al., 1987). The application of genomic data to applied fields, like biodiversity forensics, conservation, management, and restoration, for the molecular identification of unknown organismal samples, came later (*e.g.*, Forensically Informative Nucleotide Sequencing (FINS); Bartlett and Davidson (1992)). Since its inception over 20 years ago, DNA barcoding (Hebert et al., 2003a) built significantly on earlier work and has emerged as a robust method of specimen identification and species discovery across myriad multicellular eukaryotes. This endeavour has dramatically expedited millions of collected specimens to be sequenced at easily obtained short, standardized gene regions like the cytochrome *c* oxidase subunit I (5'-COI) mitochondrial locus for animals (Hebert et al., 2003b) in the largest taxon-oriented initiative to date. Despite this, DNA barcoding is not without its controversies.

The success of the single-locus approach, particularly for regulatory applications, depends crucially on two important factors: (1) the availability of high-quality specimen records found in public reference sequence databases such as the Barcode of Life Data Systems (BOLD; <http://www.barcodinglife.org>) (Ratnasingham and Hebert, 2007) and GenBank (<https://www.ncbi.nlm.nih.gov/genbank/>), and (2) the establishment of a DNA barcode gap — the notion that the maximum genetic distance observed within species is much smaller than the minimum degree of marker variation found among species (Meyer and Paulay, 2005; Meier et al., 2008). Early research has demonstrated that the presence of a DNA barcode gap hinges strongly on extant levels of species haplotype diversity gauged from

comprehensive specimen sampling at wide geographic and ecological scales (Bergsten et al., 2012; Čandek and Kuntner, 2015). Notwithstanding, many taxonomic groups lack adequate separation in their pairwise intraspecific and interspecific genetic distances due to varying rates of evolution in both genes and taxa (Pentinsaari et al., 2016). Furthermore, it has been well-demonstrated that the presence of a DNA barcode gap becomes less certain with increasing spatial scale of sampling, since interspecific distances increase due to population structuring under limited dispersal through isolation-by-distance, while intraspecific distances shrink as more closely-related species are sampled (Phillips et al., 2022). This can pose problems in cases of rare singleton species, endemics, or monotypic taxa, for instance (Ahrens et al., 2016). As a result, rapid matching of unknown samples to expertly-validated references can be compromised, leading to cases of false positives (taxon oversplitting) and false negatives (excessive lumping of taxa) due to incomplete lineage sorting, interspecies hybridization, genome introgression, species synonymy, cryptic species diversity, and misidentifications (Hubert and Hanner, 2015; Phillips et al., 2022). Another culprit includes PCR/sequencing error caused by incorrect nucleotide basecalling, chimeras/heteroplasmies, sample contamination, indels (insertion/deletions), NUMT (nuclear-mitochondrial insert) pseudogene amplification, or *Wolbachia* endosymbiont infection, among other explanations (Phillips et al., 2023). Each of these can artificially magnify levels of genetic diversity in taxon populations, rendering techniques like DNA barcoding unreliable for certain groups like marine invertebrates, which are both undersampled and possess slow rates of molecular evolution (Bucklin et al., 2011).

Recent work has argued that DNA barcoding, in its current form, is lacking in statistical rigor (Phillips et al., 2022). Most publications rely strongly on heuristic distance-based thresholds calculated from convenient specimen sample sizes (usually 5-10 specimens per species, but singleton/doubletons are more prevalent). Instead, focus should be placed on optimal sampling designs using collated taxon haplotype information, coupled with project funding and budget constraints (Phillips et al., 2015, 2019, 2020). This is necessary to infer

taxonomic identity and delineate boundaries on the basis of clustering genetic variation into putative groups reminiscent of species (Phillips et al., 2022). Of these studies, few report measures of uncertainty, such as standard errors (SEs) and confidence intervals (CIs), around estimates of intraspecific and interspecific variation (*e.g.*, Wiemers and Fiedler (2007)), calling into question the existence of a true species' DNA barcode gap (Čandek and Kuntner, 2015; Phillips et al., 2022). To support this idea, novel nonparametric locus- and species-specific metrics based on the multispecies coalescent (MSC) were recently outlined by Phillips et al. (2024) (see **Methods**).

The basic coalescent (Kingman, 1982a,b) encompasses a backwards continuous-time stochastic Markov process of allelic sampling among lineages under the infinite sites model of mutation within a large neutrally-evolving, panmictic, diploid species population of constant size (*i.e.*, Wright-Fisher dynamics) from the present into the past towards the most recent common ancestor (MRCA). Although the coalescent sampling process plays a foundational role in evolutionary biology and population genetics theory due to its inherent simplicity and flexibility (Rosenberg and Nordborg, 2002; Degnan and Rosenberg, 2009), the approach has been both under-utilized and under-appreciated as a key player within DNA barcoding (Collins and Cruickshank, 2014; Dowton et al., 2014; Hubert and Hanner, 2015; Phillips et al., 2022). Numerous theoretical and empirical investigations have shown that protein-coding sequence variation within non-recombining haploid loci like COI is generally subject to strong purifying selection and selective sweeps at nonsynonymous sites, rather than the occurrence of functionally silent substitutions within synonymous regions; consequently, DNA barcoding works in practice, but not in theory (Stoeckle and Thaler, 2014). However, because mitochondrial markers have considerably smaller effective population sizes ( $N_e$ ) compared to nuclear regions of the genome, accumulated mutations become fixed, and therefore diagnostic of species, at a much faster rate due to genetic drift (Hubert and Hanner, 2015; Phillips et al., 2022). Previously proposed MSC algorithmic approaches (of which there are too many to exhaustively list here, and which have mostly been applied to multilocus

sequence data) generally assume a molecular clock and a model of DNA sequence evolution across closely-related taxa (Ho and Duchêne, 2014; Flouri et al., 2022). From here, an estimated species phylogeny may be easily constructed (*e.g.*, with or without use of a guide tree) (*e.g.*, Rannala and Yang (2003, 2017); Yang and Rannala (2010, 2014, 2017)). Even tree- and network-based taxon delimitation methods designed for single loci, such as the Generalized Mixed Yule Coalescent (GMYC) (Pons et al., 2006; Monaghan et al., 2009; Fujisawa and Barraclough, 2013), Poisson Tree Processes (PTP) (Zhang et al., 2013; Kapli et al., 2017), and statistical parsimony/haplotype networks (*e.g.*, TCS networks (Templeton et al., 1992), haplowebs (Dellicour and Flot, 2015, 2018)), which see much use in DNA barcoding initiatives, have their flaws. Optimal performance of these methods relies heavily on careful parameter selection (*e.g.*, prior specification of branching rates in ultrametric trees in the case of GMYC) (Hubert et al., 2024) within third-party software like BEAST (Bouckaert et al., 2019), among other concerns (Talavera et al., 2013; Fonseca et al., 2021).

Hubert et al. (2024) categorized statistical species delimitation methods according to five crucial factors necessary for success: availability, reliability, scalability, understandability, and usability. Any proposed approach should score high across all of these aspects (Hubert et al., 2024). Yet, numerous tradeoffs exist, and so current methods fall short. (See Table 1 in Hubert et al. (2024) for a comprehensive overview.) Careless parameterization of established approaches by users unfamiliar with their underlying statistical characteristics may severely bias overall results in unintended ways (Hart and Sunday, 2007). In contrast, Phillips et al.’s (2024) approach is both tree- and network-free, and does not require judicious parameter setting; all that is needed is a distance matrix. Therefore, the method is easy to use and interpret, quite memory efficient, very fast to run on large datasets, and archived freely online through GitHub (<https://github.com/jphill01/DNA-Barcode-Gap-Coalescent>). Most importantly, the approach ranks highly across all criteria mentioned by Hubert et al. (2024).

The estimators from Phillips et al. (2024) represent a clear improvement over simple, yet

arbitrary, distance heuristics such as the 2% rule noted by Hebert et al. (2003a) and the 10×  
 rule (Hebert et al., 2004). These conventions form the core of threshold-based single-locus  
 species delimitation tools like Automatic Barcode Gap Discovery (ABGD) (Puillandre et al.,  
 2011), Assemble Species by Automatic Partitioning (ASAP) (Puillandre et al., 2021), and  
 the Barcode Index Number (BIN) framework (Ratnasingham and Hebert, 2013). The 2%  
 rule asserts that DNA barcodes differing by at least 2% at sequenced genomic regions should  
 be expected to originate from different biological species. In contrast, the 10× rule suggests  
 that barcode sequences displaying 10 times more genetic variation among species than within  
 taxa is suggestive of a distinct evolutionary origin. However, the lack of adoption of an  
 explicit and universally agreed-upon species concept in DNA barcoding studies is missing.  
 The ideal concept is one that readily governs lineage formation and proliferation necessary to  
 establish rigorous taxon definitions for successful delimitation of hypothesized and heuristic  
 evolutionary units using DNA barcode identification criteria (de Queiroz, 2007; Pante et al.,  
 2015; Rannala, 2015; Wells et al., 2022). Such a concept must be proposed in conjunction with  
 secondary lines of evidence (*e.g.*, morphology, ecology, geography, ontogeny, and behaviour)  
 promised by an integrative framework (Dayrat, 2005). This is the case, for instance, with  
 species existing in allopatry/sympatry, where establishment of reproductive isolation on the  
 basis of fixed diagnostic differences using the Biological Species Concept (BSC) (Mayr, 1942),  
 is challenging (Fujita et al., 2012). Notably, Fujita et al. (2012) remarks that the MSC  
 applied to the taxon delimitation problem harmonizes many species concepts since it identifies  
 independent evolutionary trajectories under the umbrella of the General Lineage Concept.  
 This makes the MSC a fitting tool to model speciation and other modes of diversification  
 (Collins and Cruickshank, 2014). In addition, the reliance on visualization approaches,  
 such as frequency histograms, dotplots, and quadrant plots, to expose DNA barcoding’s  
 limitations, has also been criticized since they can fog real species signal relative to noise  
 (Collins and Cruickshank, 2013; Phillips et al., 2022). Up until the work of Phillips et al.  
 (2024), the majority of studies (*e.g.*, Young et al. (2021)) have treated the DNA barcode

gap as a binary response. Given poor sampling depth for most taxa, a Yes/No dichotomy is inherently flawed because it can falsely imply a DNA barcode gap is present for a taxon of interest when in fact no such separation in genetic distances exists. On the other hand, in the case of sequence overlap, while the presence a single specimen record is a necessary requirement to nullify the existence of a DNA barcode gap in light of available genetic data, it is not a sufficient condition, since said taxon record could represent a species misidentification or other error. Thus, the MSC serves as an ideal model to settle ongoing debates and issues regarding DNA barcoding’s use as both a molecular identification and delineation tool. The recently introduced DNA barcode gap metrics go some way into further operationalizing this task.

Phillips et al.’s (2024) proposed statistics quantify the extent of asymmetric directionality of proportional distance distribution overlap/separation for species within well-sampled taxonomic genera based on a straightforward distance count, in a similar vein to established measures of statistical similarity such as the Kullback-Leibler (KL) divergence (Kullback and Leibler, 1951) and other related statistics. The metrics can be employed in a variety of ways, including to validate performance of marker genes for specimen identification to the species level (using nonparametric bootstrapping, as in Phillips et al. (2024)). The estimators can further be utilized to assess whether computed values are consistent with fundamental population genetic-level parameters like  $N_e$ , mutation rates ( $\mu$ ), migration rates ( $m$ ), and divergence times ( $\tau$ ), as well as derived ones, such as the population mutation rate  $\theta = 4N_e\mu$ , and the fixation index,  $F_{ST}$ . This is especially true for species under study within a statistical phylogeographic setting (Knowles and Maddison, 2002; Knowles, 2004, 2009; Mather et al., 2019). Early on, species discovery using DNA barcoding was presumed to only work for reciprocally monophyletic groups, and thus concerned itself with terminal branches of generated phylogenies rather than more basal lineages occurring deeper in hypothesized species trees (Fujita et al., 2012; Mutanen et al., 2016). Furthermore, the occurrence of short branches and large  $N_e$  within resolved phylogenies increases the likelihood of deep



202 coalescence (Degnan and Rosenberg, 2009), clouding species delimitations, which often fail or  
 203 are uncertain in broad parameter space (Carstens et al., 2013; Hickerson et al., 2006; Rannala,  
 204 2015). As DNA barcoding is a single-locus approach, it is problematic for evolutionarily young  
 205 taxa. Incomplete lineage sorting within gene genealogies, leading to the retention of ancestral  
 206 polymorphisms within populations, is a common phenomenon due to the ongoing stochastic  
 207 dynamic of mutation generating population variation, and genetic drift driving gene variants  
 208 to fixation (Degnan and Rosenberg, 2009; Rannala, 2015). The most promising way forward  
 209 in assessing the validity of Phillips et al.’s (2024) method seems to be through the use of  
 210 software such as BPP (Bayesian Phylogenetics and Phylogeography), which permits efficient  
 211 full Bayesian simulations under various MSC models (*e.g.*, MSC-I (MSC with introgression;  
 212 Flouri et al. (2020)) or MSC-M (MSC with migration; Flouri et al. (2023)), among others)  
 213 using Markov Chain Monte Carlo (MCMC) for tree parameter estimation (via the A00  
 214 option, for instance) (Flouri et al., 2018), or PHRAPL (Phylogeographic Inference using  
 215 Approximate Likelihoods) (Jackson et al., 2017a,b), which employs tractable phylogenetic  
 216 likelihood calculations via the genealogical divergence index (gdi). Other hypothesis-driven  
 217 approaches to species delimitation model selection, like Bayes Factors (BFs) (Leaché et al.,  
 218 2014), are also possible. Rigorous statistical methodology has been historically integrated  
 219 into DNA barcoding previously (Matz and Nielsen, 2005; Nielsen and Matz, 2006), though  
 220 strictly for the purpose of specimen identification. Phillips et al.’s (2024) is unique as it also  
 221 greatly extends usage to delimit species, while also being grounded strongly in evolutionary  
 222 theory.

223 While introduction of the new metrics is a step in the right direction, what appears to  
 224 be missing is a rigorous statistical treatment of the DNA barcode gap. This includes an  
 225 unbiased way to compute the statistical accuracy of Phillips et al.’s (2024) estimators arising  
 226 through problems inherent in frequentist maximum likelihood estimation for probability  
 227 distributions having bounded positive support on the closed unit interval  $[0, 1]$ . To this  
 228 end, here, a Bayesian model of the DNA barcode gap coalescent is introduced to rectify

such issues. The model allows accurate estimation of posterior means, posterior standard deviations (SDs), posterior quantiles, and credible intervals (CrIs) for the statistics given datasets of intraspecific and interspecific distances for species of interest.

## 2 Methods

### 2.1 DNA Barcode Gap Metrics

The novel nonparametric maximum likelihood estimators (MLEs) of proportional overlap/separation between intraspecific and interspecific distance distributions for a given species ( $x$ ) to aid assessment of the DNA barcode gap are as follows:

$$p_x = \frac{\#\{d_{ij} \geq a\}}{\#\{d_{ij}\}} \quad (1)$$

$$q_x = \frac{\#\{d_{XY} \leq b\}}{\#\{d_{XY}\}} \quad (2)$$

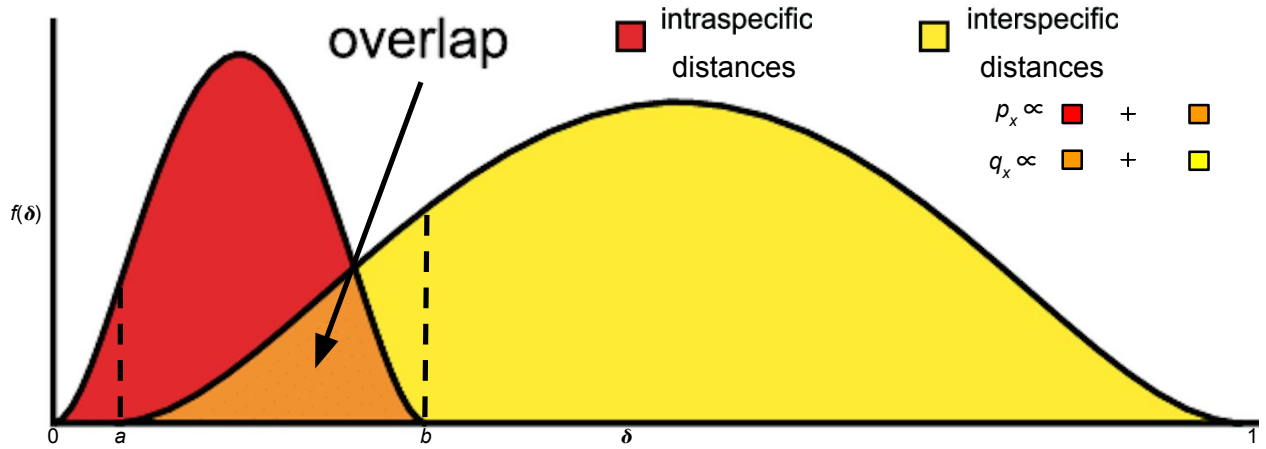
$$p'_x = \frac{\#\{d_{ij} \geq a'\}}{\#\{d_{ij}\}} \quad (3)$$

$$q'_x = \frac{\#\{d'_{XY} \leq b\}}{\#\{d'_{XY}\}} \quad (4)$$

where  $d_{ij}$  are distances within species,  $d_{XY}$  are distances among species for an *entire* genus of concern, and  $d'_{XY}$  are combined interspecific distances for a target species and its closest neighbouring species. The notation  $\#$  reflects a count. Distances form a continuous distribution and are easily computed from a model of DNA sequence evolution, such as uncorrected or corrected p-distances (Jukes and Cantor, 1969; Kimura, 1980) using, for example, the `dist.dna()` function available in the `ape` R package (Paradis et al., 2004). Quantities  $a$ ,  $a'$ , and  $b$  correspond to  $\min(d_{XY})$ ,  $\min(d'_{XY})$ , and  $\max(d_{ij})$ , the minimum interspecific distance, the minimum combined interspecific distance, and the maximum intraspecific distance, respectively (**Figure 1**). Note, distances are constrained to the interval

$[0, 1]$ , whereas the metrics are defined only on the interval  $[a/a', b]$ .

Values of the estimators obtained from Equations (1)-(4) close to or equal to zero give evidence for separation between intraspecific and interspecific distance distributions; that is, values suggest the presence of a DNA barcode gap for a target species. Conversely, values near or equal to one give evidence for distribution overlap; that is, values likely indicate the absence of a DNA barcode gap. Furthermore, calculation of the DNA barcode gap estimators is convenient as they implicitly account for total distribution area within  $[a/a', b]$  (including overlap). It should be noted however that there is a loss of information in computing estimates of Phillips et al.'s (2024) metrics since two different sets of  $(p_x, q_x)$  and  $(p'_x, q'_x)$  values cause equal but distinct underlying distance distributions of  $f_{\text{intra}}(\delta)$  and  $f_{\text{inter}}(\delta)$  (**Figure 1**). In other words, under certain circumstances, the metrics cannot be determined uniquely, and so are non-identifiable in these cases.



**Figure 1:** Modified depiction from Meyer and Paulay (2005) and Phillips et al. (2024) of the overlap/separation of intraspecific and interspecific distances ( $\delta$ ) for calculation of the DNA barcode gap metrics ( $p_x$  and  $q_x$ ) for a hypothetical species  $x$ . The minimum interspecific distance is denoted by  $a$  and the maximum intraspecific distance is indicated by  $b$ . The quantity  $f(\delta)$  is akin to a kernel density estimate of the probability density function of distances. A similar visualization can be displayed for  $p'_x$  and  $q'_x$  based on intraspecific and combined interspecific distances within the interval  $[a', b]$ .

Hence, Equations (1)-(4) are simply empirical partial means of distances falling at and below, or at and exceeding, given distribution thresholds. Notice further that  $a/a'$ , and  $b$  are also

260 the first and  $n$ th order statistics,  $X_{(1)}$  and  $X_{(n)}$ , respectively, with  $a/a' < b$ , which have been  
 261 pointed out by Phillips et al. (2022) as important for developing a statistical theory to test  
 262 for the existence of the DNA barcode gap. However, the underlying sampling distributions of  
 263  $\eta = a - b$  and  $\eta' = a' - b$  are not obvious (they involve the evaluation of convolutions). This  
 264 caveat renders a formal hypothesis test cumbersome to derive since said distributions are  
 265 not expected to be symmetric. The standard bootstrap resampling procedure (Efron, 1979)  
 266 can also fail when the estimator of interest is not well-behaved, necessitating extensions  
 267 like the  $m$ -out-of- $n$  bootstrap discussed briefly in Phillips et al. (2022). This is similar  
 268 to calculating bootstrap support values to assess how well tree nodes justify present data  
 269 (Efron et al., 1996). However, as more data are collected, support values, along with their  
 270 associated SEs, will necessarily converge to unity and zero, respectively. Phillips et al. (2024)  
 271 made an argument for the resampling of alignment sites since DNA barcodes are short in  
 272 length and sites contributing to distance distribution overlap may not be included within  
 273 generated bootstrap resamples; yet, no separation could still be evident. Equations (1)-(4)  
 274 can also be expressed in terms of empirical cumulative distribution functions (ECDFs) (see  
 275 next section). A clear connection between the ECDF and the nonparametric MLE was  
 276 noted by Efron and Tibshirani (1993) as well as others. However, calculated values are *not*  
 277 independent and identically distributed (IID) because estimated standard errors (SEs) will  
 278 depend on both the number of species sampled within the genus under study, as well as the  
 279 number of specimens sampled within a target species. To tease this out, Phillips et al. (2024)  
 280 suggests plotting estimator values against their estimated SEs, along with a simple random  
 281 downsampling scheme. In the case of two species comprising a focal genus, one well sampled  
 282 and the other poorly sampled, values of the metrics close to zero for the sufficiently sampled  
 283 species will likely possess larger SEs following downsizing to match the number of poorly  
 284 sampled specimens (Phillips et al., 2024). While this scheme appears promising, challenges  
 285 arise in uncertainty quantification as discussed later.

## 2.2 Relationship to (Multispecies) Coalescent Theory

The approach of Phillips et al. (2024) differs markedly from the traditional definition of the DNA barcoding gap laid out by Meyer and Paulay (2005) and Meier et al. (2008) in that the proposed metrics incorporate interspecific distances which *include* the target species of interest. Furthermore, if a focal species is found to have multiple nearest neighbours, then the species possessing the smallest average distance is used (Phillips et al., 2024). These schemes more accurately account for species' and coalescent histories inferred from contemporaneous samples of DNA sequences leading to instances of barcode sequence sharing, such as interspecific hybridization/introgression events (Phillips et al., 2024). For example, within Equations (3) and (4), the degree of distance distribution overlap/separation between a target taxon and its nearest neighbouring species, gauged from magnitudes of  $p'_x$  and  $q'_x$ , is directly proportional to the amount of time in which the two lineages diverged from the MRCA (Phillips et al., 2024). More specifically,  $a \propto t_{XY, \min}$  and  $b \propto t_{ij, \max}$ , the minimum and maximum (unobserved) divergence time (in  $N_e$  generations), respectively, for two different individuals ( $i, j$ ) and species ( $X, Y$ ) within a well-inventoried genus (Phillips et al., 2024). Figure 1 in Phillips et al. (2024) depicts a phylogeny for two hypothetical species,  $A$  and  $B$ , showing paraphyly based on the sequencing of two mitochondrial loci. The (unobserved) time the two species formed,  $t_{AB}$ , is intermediate between  $t_{XY, \min}$  and  $t_{ij, \max}$ , but theoretically, its maximum ( $t_{AB, \max}$ ) may occur more distantly in the past closer to the root comprising the MRCA for all sampled taxa (Phillips et al., 2024). From a coalescence perspective,  $p_x$  is the probability two lineages within species  $x$  coalesce after  $t_{XY, \min}$ , whereas  $q_x$  represents the probability a lineage within species  $x$  coalesces with a lineage in species  $k$  (within a sample) after  $t_{XY, \min}$ , but before  $t_{ij, \max}$ . This approach is similar to that of De Sanctis et al. (2022), who devised a coalescent model of taxonomic binning assignment accuracy given query and reference DNA sequences for two species found in a well-curated genomic database. However, the approach herein is inherently simpler, easily extensible, and more intuitive, since it does not make any parametric assumptions regarding mutations along branch lengths, coalescence

times, *etc.* and can be applied to more than two taxa. Thus, the quantities can be used as a criterion to assess the failure of DNA barcoding in recently radiated taxonomic groups, among other plausible biological explanations mentioned earlier.

## 2.3 The Model

### 2.3.1 Relating the DNA Barcode Gap Metrics to the Cumulative Distribution Function

Before delving into the derivation of the proposed DNA barcode gap metrics, review of some fundamental statistical theory is necessary. For a given random variable  $X$  from a specified distribution  $F$ , its cumulative distribution function (CDF) is defined by

$$F_X(t) = \mathbb{P}(X \leq t) = 1 - \mathbb{P}(X > t). \quad (5)$$

Rearranging Equation (5) gives

$$\mathbb{P}(X > t) = 1 - F_X(t) = 1 - \mathbb{P}(X \leq t), \quad (6)$$

from which it follows that

$$\mathbb{P}(X \geq t) = 1 - F_X(t) + \mathbb{P}(X = t). \quad (7)$$

The CDF has many desirable statistical properties, such as boundedness, continuity, and monotonicity. Consult any graduate-level text on mathematical statistics for reference.

Equations (1)-(4) can thus be written in terms of ECDFs via Equations (5) and (7) as follows, since the true underlying  $F$ s, are unknown *a priori*, and therefore must be estimated

328 using available sequence distance data:

$$\begin{aligned} p_x &= \mathbb{P}(d_{ij} \geq a) \\ &= 1 - \hat{F}_{d_{ij}}(a) + \mathbb{P}(d_{ij} = a) \end{aligned} \quad (8)$$

$$\begin{aligned} q_x &= \mathbb{P}(d_{XY} \leq b) \\ &= \hat{F}_{d_{XY}}(b) \end{aligned} \quad (9)$$

$$\begin{aligned} p'_x &= \mathbb{P}(d_{ij} \geq a') \\ &= 1 - \hat{F}_{d_{ij}}(a') + \mathbb{P}(d_{ij} = a') \end{aligned} \quad (10)$$

$$\begin{aligned} q'_x &= \mathbb{P}(d'_{XY} \leq b) \\ &= \hat{F}_{d'_{XY}}(b). \end{aligned} \quad (11)$$

329 Given  $n$  increasing-ordered data points, the (discrete) ECDF,  $\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{[x_i \leq t]}$ ,  
 330 comprises an increasing step function having jump discontinuities of size  $\frac{1}{n}$  at each sample  
 331 observation  $(x_i)$ , excluding ties (or steps of weight  $\frac{i}{n}$  with duplicate observations), where  
 332  $\mathbb{1}(x)$  is the indicator function. Note that  $1 - \hat{F}_n(t)$  is the complementary ECDF (CECDF),  
 333 which is a declining function. Further,  $\mathbb{P}(X = t)$  is the probability that  $X$  (a distance) takes  
 334 on the value  $t$ . The ECDF/CECDF has good asymptotic behaviour in the limit of sample  
 335 size. Equations (8)-(11) clearly demonstrate the asymmetric directionality of the proposed  
 336 metrics. Plots of the ECDFs/CECDFs for nonzero intraspecific, interspecific, and combined  
 337 interspecific distances, superimposed with corresponding values of  $a/a'$  and  $b$ , provide a new,  
 338 useful way to visualize the DNA barcode gap and Phillips et al.'s (2024). Firstly, there is no  
 339 need to tune the output (*e.g.*, number of bins and bin width in the case of histograms (Phillips  
 340 et al., 2022)). Secondly, probabilities reflecting areas under the distance curves depicted in  
 341 **Figure 1** can be directly read off the ECDF/CECDF curves. Values of the metrics can  
 342 also be determined from direct inspection of the ECDF/CECDF plot by determining the

corresponding  $y$ -value (cumulative probability) for a given value of  $x$  (distance).

### 2.3.2 Integrating Frequentist and Bayesian Inference

A major criticism of large sample (frequentist) theory is that it relies on asymptotic properties of the MLE (whose population parameter is assumed to be a fixed but unknown quantity), such as estimator normality and consistency as the sample size approaches infinity. This problem is especially pronounced in the case of binomial proportions (Newcombe, 1998). The estimated Wald SE of the sample proportion, is given by  $\widehat{SE}[\hat{p}] = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ , where  $\hat{p} = \frac{Y}{n}$  is the MLE,  $Y$  is the total number of successes ( $Y = \sum_{i=1}^n y_i$ ), and  $n$  is the total number of trials (*i.e.*, sample size). However, the above formula for the standard error is problematic for several reasons. First, it is a Normal approximation which makes use of the central limit theorem (CLT); thus, large sample sizes are required for reliable estimation. When few observations are available, SEs will be large and inaccurate, leading to low statistical power to detect a true DNA barcode gap when one actually exists. Further, resulting interval estimates could span values less than zero or greater than one, or have zero width, which is practically meaningless. Second, when proportions are exactly equal to zero or one, resulting SEs will be exactly zero, rendering  $\widehat{SE}[\hat{p}]$  given above completely useless. In the context of the proposed DNA barcode gap metrics, values obtained at the boundaries of their support are often encountered. Therefore, reliable calculation of SEs is not feasible. Given the importance of sufficient sampling of species genetic diversity for DNA barcoding initiatives, a different statistical estimation approach is necessary.

Bayesian inference offers a natural path forward in this regard since it allows for straightforward specification of prior beliefs concerning unknown model parameters and permits the seamless propagation of uncertainty, when data are lacking and sample sizes are small, through integration with the likelihood function associated with true generating processes. The posterior distribution ( $\pi(\theta|Y)$ ) is given by Bayes' theorem up to a proportionality  $\pi(\theta|Y) \propto \pi(Y|\theta)\pi(\theta)$ , where  $\theta$  are unobserved parameters,  $Y$  are known



data,  $\pi(Y|\theta)$  is the likelihood, and  $\pi(\theta)$  is the prior. As a consequence, because parameters are treated as random variables, Bayesian models are much more flexible and generally more easily interpretable compared to frequentist approaches. Under the Bayesian paradigm, entire posterior distributions, along with their summaries (*e.g.*, CrIs) are outputted, rather than just long run behaviour reflected in sampling distributions, p-values, and CIs as in the frequentist case, thus allowing direct probability statements to be made.

### 2.3.3 Relating the DNA Barcode Gap Metrics to Mixture Models

Essentially, from a statistical perspective, the goal herein is to nonparametrically estimate probabilities corresponding to extreme tail quantiles for positive highly skewed distributions on the unit interval (or any closed subinterval thereof). Here, it is sought to numerically approximate the extent of proportional overlap/separation of intraspecific and interspecific distance distributions within the subinterval  $[a/a', b]$ . This is a challenging computational problem within the current study as detailed in subsequent sections. The usual approach employs kernel density estimation (KDE), along with numerical or Monte Carlo integration of complicated probability distribution functions (PDFs), and invocation of extreme value theory (EVT) to calculate overlap (such as that for species with unusually large distances suggestive of cryptic species diversity, or deep divergence due to long periods of genetic isolation); however, this requires careful selection of the bandwidth parameter, which would need to be identical for both distributions, among other considerations. This becomes problematic when fitting finite mixture models using MCMC, where nonidentifiability is rampant due to the occurrence of label switching, making clusters hard to discern across sampling iterations, and model selection challenging due to lack of exchangeability among observations (Stephens, 2000). For DNA barcode gap estimation, this would correspond to a two-component mixture (one for intraspecific distance comparisons, and the other for interspecific comparisons), with one or more distribution modes or curve intersection points between components, and the presence of zero distance inflation. This makes parameter

estimation difficult using methods like the Expectation-Maximization (EM) algorithm (Dempster et al., 1977) as the algorithm may become stuck in suboptimal regions of the parameter search space and prematurely converge to local optima.

The abovementioned statistical issues seems wholly plausible, especially given that the genus-level distribution of genetic distances is a mixture of closely-related (*i.e.*, nearest neighbour) and more distant species. Examining **Figure 1**, depending on divergence times and other elements, intraspecific and interspecific distributions would be expected to comprise greater density in the upper (right) tails (which make up more distantly-related taxa relative to a target species) compared to the lower (left) tails (which contain only the closest conspecifics) since there are more combinations of distance comparisons possible. A key problem affecting reliable inference of the DNA barcode gap metrics is undersampling of genus-level and/or interspecific variation. An immediate, but non-trivial solution appears to be a move toward next-generation DNA barcoding incorporating multiple loci (Collins and Cruickshank, 2014; Dowton et al., 2014). In theory, this would facilitate the detection of missing operational taxonomic units (OTUs), both extant and extinct, when attempting to better characterize a genus. In this context, Phillips et al.’s (2024) statistics could be informative in this regard for studies of human phylogeography and ancestry composition where historical admixture between sympatric populations was commonplace (*e.g.*, modern human/Denisova interbreeding). Here, for simplicity, an alternate route is taken to avoid these obstacles.

#### 2.3.4 Bayesian Model Setup

Counts,  $y$ , of overlapping distances (as expressed in the numerator of Equations (1)-(4)) are treated as binomially distributed; that is,  $\pi(y|\theta) = \binom{k}{y}\theta^y(1 - \theta)^{k-y}$ , where  $\binom{k}{y}$  is the binomial coefficient expressing the number of ways to choose  $y$  elements from  $k$  total objects. Total counts have expectation  $\mathbb{E}[Y|\theta] = k\theta$ , and variance  $\mathbb{V}[Y|\theta] = k\theta(1 - \theta)$ , where  $k = \{N, C\}$  are total count vectors of intraspecific and combined interspecific distances,

421 respectively, for a target species along with its nearest neighbour species, and  $k = M$  is a  
 422 total count vector for all interspecific species comparisons. This follows from the fact that  
 423 the ECDF/CECDF is binomially distributed. The quantity thus being estimated is the  
 424 parameter vector  $\underline{\theta} = \{p_x, q_x, p'_x, q'_x\}$ . The metrics encompassing  $\underline{\theta}$  are presumed to follow a  
 425 Beta( $\alpha, \beta$ ) distribution, with real shape parameters  $\alpha$  and  $\beta$ , which is a natural choice of  
 426 prior on probabilities. That is,  $\pi(\theta|\alpha, \beta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ , where  $B(\alpha, \beta)$  is the beta  
 427 function evaluated at  $\alpha$  and  $\beta$ . The beta distribution has a prior mean of  $\mathbb{E}[\theta|\alpha, \beta] = \frac{\alpha}{\alpha+\beta}$  and  
 428 a prior variance equal to  $\mathbb{V}[\theta|\alpha, \beta] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ . In the case where  $\alpha = \beta$ , all generated  
 429 Beta( $\alpha, \beta$ ) distributions will possess the same prior expectation, whereas the prior variance  
 430 will shrink as both  $\alpha$  and  $\beta$  increase. Such a scheme is quite convenient since the beta  
 431 distribution is conjugate to the binomial distribution. Thus, the posterior distribution is also  
 432 beta distributed, specifically, Beta( $\alpha + Y, \beta + n - Y$ ), having expectation  
 433  $\mathbb{E}[\theta|Y, \alpha, \beta] = \frac{\alpha+Y}{\alpha+\beta+n}$  and variance  $\mathbb{V}[\theta|Y, \alpha, \beta] = \frac{(\alpha+Y)(\beta+n-Y)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$ . In the context of DNA  
 434 barcoding, it is important that the DNA barcode gap metrics effectively differentiate between  
 435 extremes of no overlap/complete separation and full overlap/no separation,  
 436 corresponding to values of the metrics equal to 0 and 1 (equivalent to total distance counts  
 437 of 0 and  $n$ ), respectively. These extremes yield a posterior expectation of  
 438  $\mathbb{E}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha}{\alpha+\beta+n}$  and a posterior variance of  $\mathbb{V}[\theta|Y = 0, \alpha, \beta] = \frac{\alpha(\beta+n)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$   
 439 and  $\mathbb{E}[\theta|Y = n, \alpha, \beta] = \frac{\alpha+n}{\alpha+\beta+n}$  and  $\mathbb{V}[\theta|Y = n, \alpha, \beta] = \frac{(\alpha+n)\beta}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$ . Note, the posterior  
 440 variances are equivalent at these thresholds for all  $\alpha = \beta$ .

441 Parameters were given an uninformative Beta(1, 1) prior, which is identical to a standard  
 442 uniform (Uniform(0, 1)) prior since it places equal probability on all parameter values within  
 443 its support. This distribution has an expected value of  $\mu = \frac{1}{2} = 0.500$  and a variance of  
 444  $\sigma^2 = \frac{1}{12} \approx 0.083$ . Further, the posterior is Beta( $Y+1, n-Y+1$ ), from which various moments  
 445 such as the expected value  $\mathbb{E}[\theta|Y] = \frac{Y+1}{n+2}$  and variance  $\mathbb{V}[\theta|Y] = \frac{(Y+1)(n-Y+1)}{(n+2)^2(n+3)}$ , and other  
 446 quantities, can be easily calculated. Clearly,  $\mathbb{E}[\theta|Y = 0] = \frac{1}{n+2}$  and  $\mathbb{V}[\theta|Y = 0] = \frac{n+1}{(n+2)^2(n+3)}$ ,  
 447 and  $\mathbb{E}[\theta|Y = n] = \frac{n+1}{n+2}$  and  $\mathbb{V}[\theta|Y = n] = \frac{n+1}{(n+2)^2(n+3)}$ . For example, with a single distance

448 ( $n = 1$ ) for a given species (wherein two specimens are compared), and where the distance  
 449 is not overlapping, on average,  $\theta$  will be equal to  $\mu = \frac{1}{3} = 0.333$ , with a variance of  
 450  $\sigma^2 = \frac{2}{36} = \frac{1}{18} \approx 0.056$ . In contrast, with  $n = 100$  individuals and there is no overlap, the  
 451 mean reduces to  $\mu = \frac{1}{102} \approx 0.001$ , with a variance of  $\sigma^2 = \frac{101}{1071612} = 9.425 \times 10^{-5}$ . When  
 452  $n = 1$  and there is overlap, the mean is  $\mu = \frac{2}{3} = 0.667$ , having a variance of  $\sigma^2 \approx 0.056$ . For  
 453  $n = 100$  where all distances overlap,  $\theta$  has expectation  $\mu = \frac{101}{102} \approx 0.990$  and variance  
 454  $\sigma^2 \approx 9.425 \times 10^{-5}$ . These edge case examples have good asymptotic behaviour with  
 455 increasing sample size (variance tends to zero), and demonstrate that the Beta(1, 1)  
 456 distribution serves as a reasonable first-pass prior for DNA barcode gap estimation using  
 457 Phillips et al.'s (2024) metrics, though tighter bounds can likely be achieved. In general  
 458 however, when possible, it is always advisable to incorporate prior information, even if only  
 459 weak, rather than simply imposing complete ignorance in the form of a flat prior distribution.  
 460 In the case of unimodal distributions, the (estimated) posterior mean often possesses the  
 461 property that it readily decomposes into a convex linear combination, in the form of a  
 462 weighted sum, of the (estimated) prior mean and the MLE. That is  
 463  $\hat{\mu}_{\text{posterior}} = w\hat{\mu}_{\text{prior}} + (1 - w)\hat{\mu}_{\text{MLE}}$ , where for the beta distribution,  $w = \frac{\alpha + \beta}{\alpha + \beta + n}$ . Therefore,  
 464 with sufficient data,  $w \rightarrow 0$  as  $n \rightarrow \infty$ , regardless of the values of  $\alpha$  and  $\beta$ , and the choice  
 465 of prior distribution becomes less important since the posterior will be dominated by the  
 466 likelihood. For the Beta(1, 1),  $w = \frac{2}{2+n}$ , with  $n = 1$  giving  $w = \frac{2}{3} = 0.667$ . For  $n = 100$ , the  
 467 prior weight is  $w = \frac{2}{102} \approx 0.0196$ . The full Bayesian model for species  $x$  is thus given by

$$\begin{aligned}
 y_{\text{lwr}} &\sim \text{Binomial}(N, p_{\text{lwr}}) \\
 y_{\text{upr}} &\sim \text{Binomial}(M, p_{\text{upr}}) \\
 y'_{\text{lwr}} &\sim \text{Binomial}(N, p'_{\text{lwr}}) \\
 y'_{\text{upr}} &\sim \text{Binomial}(C, p'_{\text{upr}}) \\
 p_{\text{lwr}}, p_{\text{upr}}, p'_{\text{lwr}}, p'_{\text{upr}} &\sim \text{Beta}(1, 1)
 \end{aligned} \tag{12}$$

Note that  $p_x$ ,  $q_x$ ,  $p'_x$ , and  $q'_x$  in Equations (1)-(4) are denoted  $p_{\text{lwr}}$ ,  $p_{\text{upr}}$ ,  $p'_{\text{lwr}}$ , and  $q'_{\text{upr}}$ , respectively within Equation (12) for easy distinction between MLEs and Bayesian posterior estimates. Further,  $y_{\text{lwr}}$ ,  $y_{\text{upr}}$ ,  $y'_{\text{lwr}}$ , and  $y'_{\text{upr}}$  are calculated from the numerators of Equations (1)-(4), respectively. The above statistical theory and derivations lay a good foundation for the remainder of this paper.

### 2.3.5 Computational Details

The proposed model is inherently vectorized to allow processing of multiple species datasets simultaneously. Model fitting was achieved using the Stan probabilistic programming language (Carpenter et al., 2017) framework for Hamiltonian Monte Carlo (HMC) (Neal, 2011) via the No-U-Turn Sampler (NUTS) sampling algorithm (Hoffman and Gelman, 2014) through the `rstan` R package (version 2.32.6) (Stan Development Team, 2023) in R (version 4.4.1) (R Core Team, 2024). Four Markov chains were run for 2000 iterations each in parallel across four cores with random parameter initializations. Within each chain, a total of 1000 samples was discarded as warmup (*i.e.*, burnin) to reduce dependence on starting conditions and to ensure posterior samples are reflective of the equilibrium distribution. Further, 1000 post-warmup draws were utilized per chain during the sampling phase. Because HMC/NUTS results in dependent samples that are minimally autocorrelated, chain thinning is not required. Each of these tuning parameters reflect default MCMC settings in Stan to control both bias and variance, respectively, in the resulting draws. All analyses in the present work were carried out on a 2023 Apple MacBook Pro with M2 chip and 16 GB RAM running macOS Ventura 13.2. A random seed was set to ensure reproducibility of model results. Outputted estimates were rounded to three decimal places of precision. Posterior draws and posterior distributions of the metrics were visualized as scatterplots and KDE plots using the `ggplot()` function within the `ggplot2` R package (version 3.5.1) (Wickham, 2016) with the default Gaussian kernel and optimal smoothness selection. To successfully run the Stan program, end users must have installed an appropriate compiler (such as GCC

or Clang) which is compatible with their operating system, such as macOS.

Convergence was assessed both visually and quantitatively as follows: (1) through examining parameter traceplots, which depict the trajectory of accepted MCMC draws as a function of the number of iterations, (2) through monitoring the Gelman-Rubin potential scale reduction factor statistic ( $\hat{R}$ ) (Gelman and Rubin, 1992; Vehtari et al., 2021), which measures the concordance of within-chain *versus* between-chain variance, and (3) through calculating the effective sample size (ESS) for each parameter, which quantifies the number of independent samples generated Markov chains are equivalent to. Mixing of chains was deemed sufficient when traceplots looked like “fuzzy caterpillars”,  $\hat{R} < 1.01$ , and effective sample sizes were reasonably large (Gelman et al., 2020). After sampling, a number of summary quantities were reported, including posterior means, posterior SDs, and posterior quantiles from which 95% CrIs could be computed to make probabilistic inferences concerning true population parameters. To validate the overall correctness of the proposed statistical model given by Equation (12), as a means of comparison, posterior predictive checks (PPCs) were also employed to generate binomial random variates in the form of mean counts from the posterior predictive distribution; that is  $\underline{\gamma} = \{Np_x, Mq_x, Np'_x, Cq'_x\}$  to verify that the model adequately captures relevant features of the observed data.

### 2.3.6 Probabilistic Interpretation of the DNA Barcode Gap Metrics

The proposed Bayesian model outlined here has a straightforward interpretation (**Table 1**).

**Table 1:** Interpretation of the DNA barcode gap estimators within  $[a/a', b]$ 

| Parameter              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_x/p_{\text{lwr}}$   | When $p_x/p_{\text{lwr}}$ is close to 0 (1), it suggests that the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average, while the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average; that is, there is (no) evidence for a DNA barcode gap.                                                                                                                                                                                                                                                                                                           |
| $q_x/p_{\text{upr}}$   | When $q_x/p_{\text{upr}}$ is close to 0 (1), it suggests that the probability of interspecific (intraspecific) distances being larger (smaller) than intraspecific (interspecific) distances is high (low) on average, while the probability of intraspecific (interspecific) distances being larger (smaller) than interspecific (intraspecific) distances is low (high) on average; that is, there is (no) evidence for a DNA barcode gap.                                                                                                                                                                                                                                                                                                           |
| $p'_x/p'_{\text{lwr}}$ | When $p'_x/p'_{\text{lwr}}$ is close to 0 (1), it suggests that the probability of intraspecific (combined interspecific distances for a target species and its nearest neighbour species) distances being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average, while the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average; that is, there is (no) evidence for a DNA barcode gap. |
| $q'_x/p'_{\text{upr}}$ | When $q'_x/p'_{\text{upr}}$ is close to 0 (1), it suggests that the probability of combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) being larger (smaller) than intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) is high (low) on average, while the probability of intraspecific distances (combined interspecific distances for a target species and its nearest neighbour species) being larger (smaller) than combined interspecific distances for a target species and its nearest neighbour species (intraspecific distances) is low (high) on average; that is, there is (no) evidence for a DNA barcode gap. |

## 2.4 *Agabus* Case Study

The DNA barcode gap statistics have been shown to hold strong promise for reliable DNA barcode gap assessment when applied to the predatory diving beetles *Agabus* (Coleoptera: Dytiscidae) (Phillips et al., 2024). With a Holarctic range, despite their ease of sampling

and well-established taxonomy, this group possesses few morphologically-distinct taxonomic characters that readily facilitate their assignment to the species level (Bergsten et al., 2012). Accurate taxon discrimination within *Agabus* is paramount since several species are currently classified as endangered on the International Union for Conservation of Nature (IUCN) Red List (<https://www.iucnredlist.org>) due to habitat fragmentation/loss, water pollution, and climate change. Further, the proposed metrics indicate that sister species pairs from this taxon are often difficult to distinguish on the basis of their DNA barcode sequences (Phillips et al., 2024), which may be caused by *Agabus*'s increased degree of non-monophyly at wider geospatial scales, occurrence of hybridization/introgression, or presence of synonymy (Bergsten et al., 2012). Using sequence data from three mitochondrial cytochrome markers (5'-COI, 3'-COI, and cytochrome *b* (CYTB)) obtained from BOLD and GenBank, results from Phillips et al. (2024) highlight that DNA barcoding has been a one-sided argument. Phillips et al.'s (2024) findings point to the need to balance both the sufficient collection of specimens, as well as the extensive sampling of species. Not surprisingly, DNA barcode libraries for most taxonomic groups are biased toward the latter effort (Phillips et al., 2015, 2019, 2022).

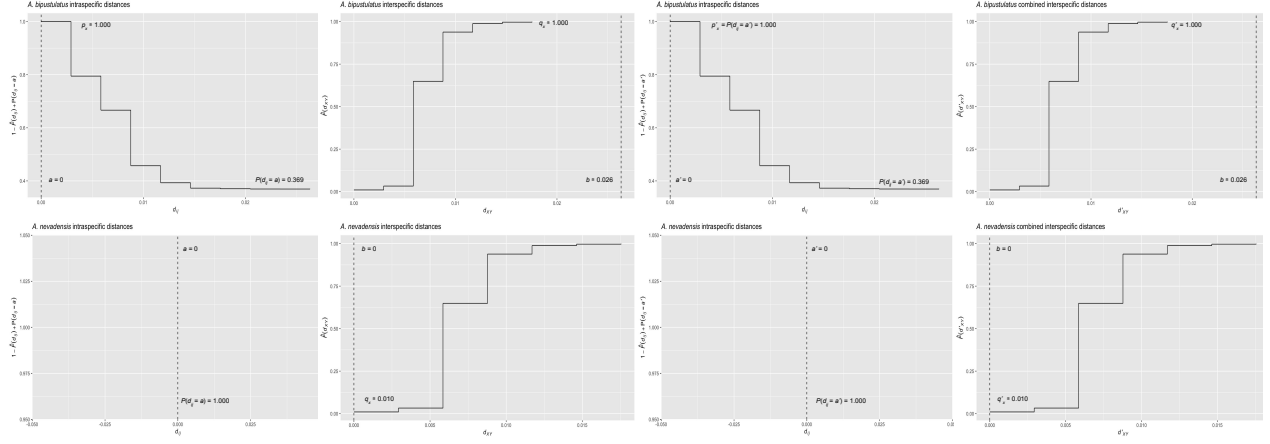
The *Agabus* CYTB dataset analyzed by Phillips et al. (2024) is revisited herein for the species *A. bipustulatus* and *A. nevadensis*, since these taxa were the sole representatives for this locus, with the most and the least specimen records, respectively ( $N = 701$  and  $N = 2$ ) across all three assessed molecular markers. Briefly, using the R package **MACER** (Young et al., 2021), DNA sequences were downloaded from GenBank and BOLD and processed using a stringent workflow to obtain a 343-bp high-quality FASTA alignment representing 46 unique haplotypes. The resulting alignment contained no stop codons, ambiguous nucleotides, or other problematic artifacts that bias genetic diversity estimates. Further, both species- and genus-level distance outliers were excluded based on the interquartile range (IQR) (Young et al., 2021). Genetic distances were calculated using uncorrected p-distances. *A. bipustulatus* comprised 45 total haplotypes, whereas



*A. nevadensis* possessed only one haplotype. Note, DNA barcode gap estimation is only possible for species having at least two specimen records because only one distance can be calculated. This dataset is a prime illustrative example highlighting the issue of inadequate taxon sampling, which arises frequently in large-scale phylogenetic and phylogeographic studies, in several respects. First, from a statistical viewpoint, sample sizes reflect extremes in reliable parameter estimation. Second, from a DNA barcoding perspective, *Agabus* currently comprises about 200 extant Linnean species according to the Global Biodiversity Information Facility (GBIF) (<https://www.gbif.org>); yet, due to the level of convenience sampling inherent in taxonomic collection efforts for this genus, adequate representation of species and genetic diversity is far from complete. Indeed, herein, species coverage for CYTB is only around 1.000%. Not surprisingly, CYTB performed poorly as a DNA barcode locus for *Agabus* despite having the highest specimen representativeness for a given species ( $N = 701$ , *A. bipustulatus*) and displaying good separation of genetic sequences among all three mitochondrial genes examined in Phillips et al.'s (2024) study; yet no barcode gap was observed since all metrics were found to be equal to 1.000.

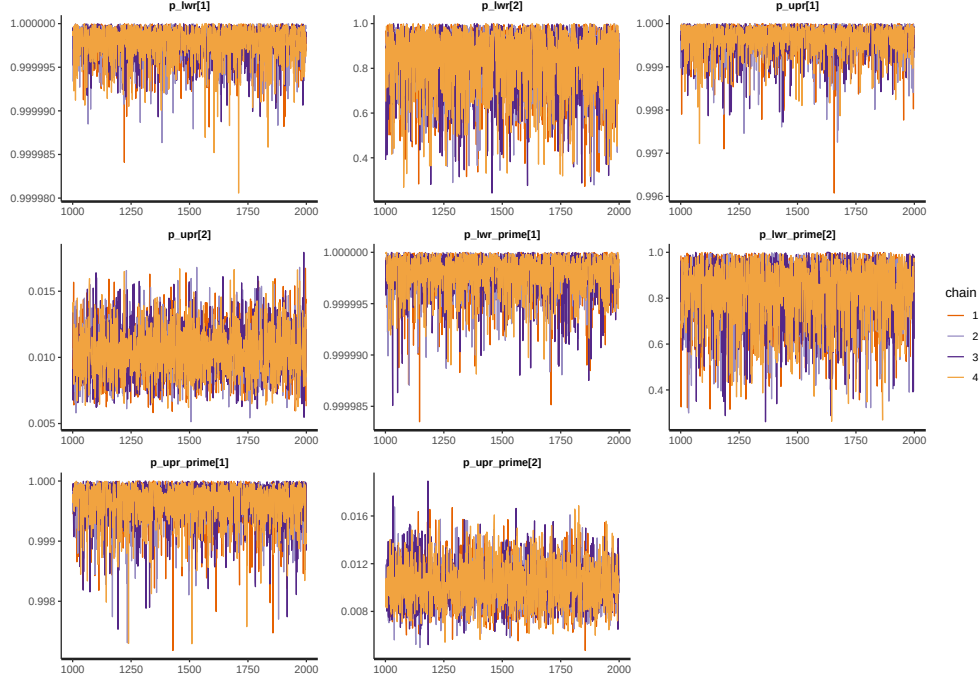
### 3 Results and Discussion

ECDF/CECDF plots were generated using the base R function `ecdf()`, along with `ggplot()`, and are displayed in **Figure 2**. Interpretation is simple: in the case of *A. bipustulatus*,  $p_x$ , all distances are greater than  $a = 0$ , with 36.900% of distances equal to  $a$ . Conversely, both  $q_x$  and  $q'_x$  equal one because all interspecific distances are less than  $b \approx 0.026$ ; for *A. nevadensis*, both metrics are approximately equal to 0.010 since only about 1% of distances fall below  $b = 0$ .



**Figure 2:** ECDF/CECDF plots of the DNA barcode gap metrics for *A. bipustulatus* and *A. nevadensis* with  $a/a'$  and  $b$  clearly indicated as dashed vertical lines. When all genetic distances are equal to zero, no step function is shown.

MCMC parameter traceplots showed rapid mixing of chains to the stationary distribution (Figure 3). Further, all  $\hat{R}$  and ESS values (both not shown) were close to their recommended cutoffs of one and thousands of samples, respectively, indicating chains are both well-mixed and have converged to the posterior distribution. Bayesian posterior estimates were reported alongside frequentist MLEs, in addition to SEs, posterior SDs, 95% CIs, and 95% CrIs (Table 2).



**Figure 3:** MCMC parameter traceplots applied to *A. bipustulatus* ([1];  $N = 701$ ) and *A. nevadensis* ([2];  $N = 2$ ) for CYTB across 1000 post-warmup iterations.

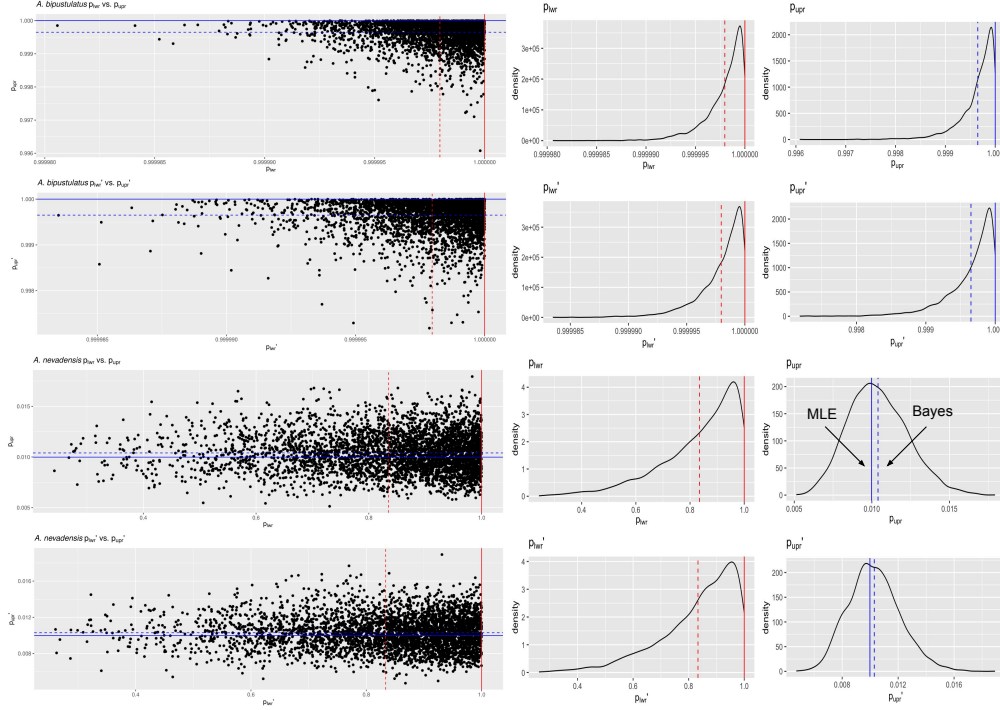
**Table 2:** Nonparametric frequentist and Bayesian estimates of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ( $N = 701$ ) and *A. nevadensis* ( $N = 2$ ) for CYTB, including 95% CIs and CrIs. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

| Species                | Parameter       | MLE (SE, 95% CI)            | Bayes Est. (SD; 95% CrI)   |
|------------------------|-----------------|-----------------------------|----------------------------|
| <i>A. bipustulatus</i> | $p_x/p_{lwr}$   | 1.000 (0.000; 1.000-1.000)  | 1.000 (0.000; 1.000-1.000) |
| <i>A. bipustulatus</i> | $q_x/p_{upr}$   | 1.000 (0.000; 1.000-1.000)  | 1.000 (0.000; 0.999-1.000) |
| <i>A. bipustulatus</i> | $p'_x/p'_{lwr}$ | 1.000 (0.000; 1.000-1.000)  | 1.000 (0.000; 1.000-1.000) |
| <i>A. bipustulatus</i> | $q'_x/p'_{upr}$ | 1.000 (0.000; 1.000-1.000)  | 1.000 (0.000; 0.999-1.000) |
| <i>A. nevadensis</i>   | $p_x/p_{lwr}$   | 1.000 (0.000; 1.000-1.000)  | 0.835 (0.144; 0.470-0.996) |
| <i>A. nevadensis</i>   | $q_x/p_{upr}$   | 0.010 (0.002; 0.006-0.014)  | 0.010 (0.002; 0.007-0.014) |
| <i>A. nevadensis</i>   | $p'_x/p'_{lwr}$ | 1.000 (0.000; 1.000-1.000)  | 0.834 (0.138; 0.481-0.994) |
| <i>A. nevadensis</i>   | $q'_x/p'_{upr}$ | 0.010 (0.070; -0.128-0.148) | 0.010 (0.002; 0.007-0.014) |

573 CIs were calculated using the usual large sample  $(1 - \alpha)100\%$ -level interval estimate given  
574 by  $\hat{p} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ , where  $z_{1-\frac{\alpha}{2}} = 1.960$  is the critical value for 95% confidence (*i.e.*,  
575 the 97.5th percent quantile from the standard Normal distribution), and  $\alpha$  is the stated  
576 significance level (here, 5%). Given a  $(1 - \alpha)100\%$  CI, with repeated sampling, on average

$(1 - \alpha)100\%$  of constructed intervals will contain the true parameter of interest, assuming a  
correctly-specified statistical model; on the other hand, any given CI will either capture or  
exclude the true parameter with 100% certainty. This in stark contrast to a CrI, where the  
true parameter is contained within said interval with  $(1 - \alpha)100\%$  probability, contingent  
on the data generating model being correct. Note, by default Stan computes equal-tailed  
(central) CrIs such that there is equal area situated in the left and right tails of the posterior  
distribution. For a 95% CrI, this corresponds to the 2.5th and 97.5th percent quantiles.  
However, constructed intervals are usually only valid for symmetric or nearly symmetric  
distributions. Given the bounded nature of the DNA barcode gap metrics, whose posterior  
distributions, as expected, show considerable skewness, an alternative approach to reporting  
CrIs, such as Highest Posterior Density (HPD) intervals (Chen and Shao, 1999) or shortest  
probability intervals (SPIn) (Liu et al., 2015) is warranted. Although the data for *Agabus*  
were not found to be consistent with a value of zero for the metrics (suggesting the existence  
of a DNA barcode gap) for either species, this is clearly a case that must differentiated from  
one where some or all of the estimators attain a value of one (indicating no DNA barcode gap  
likely exists). As such asymmetric intervals generally attain greater statistical efficiency (in  
the form of smaller Mean Squared Error (MSE) or variance) and higher coverage probabilities  
than more standard interval estimates, careful in-depth comparison is left for future work.

Findings based on nonparametric MLEs and Bayesian posterior means were quite  
comparable with one another and show evidence of complete overlap in intraspecific,  
interspecific, and combined interspecific distances for *A. bipustulatus* in both the  
 $p/q/p_{\text{lwr}}/p_{\text{upr}}$  and  $p'/q'/p'_{\text{lwr}}/p'_{\text{upr}}$  directions since the metrics and their intervals attain  
magnitudes very close or equal to one, with minimal noise (**Table 2**). As a result, this  
likely indicates that no DNA barcode gap is present for this species. Such findings are strongly  
reinforced by the very tight clustering of posterior draws (**Figure 4**) and associated interval  
estimates owing to the large number of specimens sampled for this species.



**Figure 4:** Scatterplots (black solid points) and distributions (black solid lines) depicting the DNA barcode gap metrics for *A. bipustulatus* ( $N = 701$ ) and *A. nevadensis* ( $N = 2$ ) across CYTB based on 4000 Bayesian posterior draws. MLEs and posterior means are displayed as coloured (red/blue) solid and dashed lines for the metrics, respectively.

On the other hand, the situation for *A. nevadensis* is more nuanced, as posterior values are further spread out (**Table 2** and **Figure 4**), suggesting less overall certainty in true parameter values given the low specimen sampling coverage for this taxon. Of note, calculated SEs and SDs are large, with the 95% CIs and 95% CrIs being quite wide in most cases, consistent with much uncertainty regarding the computed frequentist and Bayesian posterior means of the DNA barcode gap metrics. For instance, the Bayesian analysis suggests that the data are consistent with both  $p_{\text{lowr}}$  and  $p'_{\text{lowr}}$  ranging from approximately 0.500-1.000 due to the large SD of about 0.140. The posterior means associated with these estimates themselves are also far from one. The CrI for  $p_{\text{upr}}$  and  $p'_{\text{upr}}$  also spans an order of magnitude, with the lower tail estimate of 0.007 being very close to zero. Further, regarding the frequentist analysis for the same species, the 95% CI for  $q_x$  is quite wide, reflecting considerable uncertainty in its true parameter value. Similarly, that for  $q'_x$  extends to negative values at the left endpoint,

due to the corresponding SE of 0.070 being too high as a result of the extremely low sample size of  $n = 2$  individuals sampled (**Table 2**). Since the 95% CI for  $q'_x$  truncated at the lower endpoint includes the value of zero, the null hypothesis for the presence of a DNA barcode gap cannot be rejected. Despite this, it is worth noting that truncation is not standard statistical practice and will likely lead to an interval with less than 95% nominal coverage having high relative error. In such cases, more appropriate confidence interval methods like the Wilson score interval, the exact (Clopper-Pearson) interval, or the Agresti-Coull interval should be employed (Newcombe, 1998; Agresti and Coull, 1998). KDEs for *A. bipustulatus* are strongly left (negatively) skewed (**Figure 4**), whereas those for *A. nevadensis* exhibit more symmetry, especially for  $p_{\text{upr}}$  and  $p'_{\text{upr}}$  (**Figure 4**). These differences are likely due to the stark contrast in sample sizes for the two examined species. Nevertheless, simulated counts of overlapping specimen records from the posterior predictive distribution (**Table 3**) were found to be very close to observed counts for both species, indicating that the proposed model adequately captures underlying variation.

**Table 3:** Posterior predictive checks of distance distribution overlap/separation for the DNA barcode gap coalescent model parameters applied to *A. bipustulatus* ( $N = 701$ ) and *A. nevadensis* ( $N = 2$ ) for CYTB. CrIs are based on 4000 posterior draws. All parameter estimates are reported to three decimal places of precision.

| Species                | Variable          | $Y$        | $n$        | Bayes Est. (SD; 95% CrI)                  |
|------------------------|-------------------|------------|------------|-------------------------------------------|
| <i>A. bipustulatus</i> | $y_{\text{lwr}}$  | 491401.000 | 491401.000 | 491400.018 (1.378; 491396.000-491401.000) |
| <i>A. bipustulatus</i> | $y_{\text{upr}}$  | 2804.000   | 2804.000   | 2803.019 (1.433; 2799.000-2804.000)       |
| <i>A. bipustulatus</i> | $y'_{\text{lwr}}$ | 491401.000 | 491401.000 | 491400.008 (1.412; 491396.000-491401.000) |
| <i>A. bipustulatus</i> | $y'_{\text{upr}}$ | 2804.000   | 2804.000   | 2802.992 (1.429; 2799.000-2804.000)       |
| <i>A. nevadensis</i>   | $y_{\text{lwr}}$  | 4.000      | 4.000      | 3.355 (0.888; 1.000-4.000)                |
| <i>A. nevadensis</i>   | $y_{\text{upr}}$  | 28.000     | 2804.000   | 29.151 (7.620; 16.000-45.000)             |
| <i>A. nevadensis</i>   | $y'_{\text{lwr}}$ | 4.000      | 4.000      | 3.325 (0.884; 1.000-4.000)                |
| <i>A. nevadensis</i>   | $y'_{\text{upr}}$ | 28.000     | 2804.000   | 28.942 (7.409; 15.000-45.000)             |

Obtained results suggest that use of the Beta(1, 1) prior may not be appropriate given a low number of collected individuals for most taxa in DNA barcoding efforts. This suggests that further consideration of more informative beta priors is worthwhile. This is clear for *A. nevadensis*, where use of a stronger prior will likely eliminate the issue regarding accurate estimation of  $p_{\text{lwr}}$  and  $p'_{\text{lwr}}$  and stabilize their uncertainty.

## 4 Conclusion

Herein, the accuracy of the DNA barcode gap was analyzed from a rigorous statistical lens to expedite both the curation and growth of reference sequence libraries, ensuring they are populated with high quality, statistically defensible specimen records fit for purpose to address standing questions in ecology, evolutionary biology, biodiversity forensics, conservation, management, and restoration. To accomplish this, recently proposed, easy to calculate, nonparametric MLEs were formally derived using ECDFs/CECDFs and applied to assess the extent of overlap/separation of distance distributions within and among two species of predatory water beetles in the genus *Agabus* sequenced at CYTB using a Bayesian binomial count model with conjugate beta priors. Findings highlight a high level of parameter uncertainty for *A. nevadensis*, whereas posterior estimates of the DNA barcode gap metrics for *A. bipustulatus* are much more certain. Based on these results, it is imperative that specimen sampling be prioritized to better reflect actual species boundaries. More generally, apart from the metrics being employed to better highlighting the importance of within-species genetic diversity versus between-species divergence, it is expected that the approach developed herein will be of broad utility in applied fields, such as DNA-based detection of seafood fraud within global supply chains, and in the determination of species occupancy/detection probabilities at ecological sites of interest using active and passive environmental DNA (eDNA) methods such as metabarcoding.

Since the DNA barcode gap metrics often attain values very close to zero (suggesting no overlap and complete separation of distance distributions) and/or very near one (indicating no separation and complete overlap), in addition to more intermediate values, a noninformative Beta( $\frac{1}{2}, \frac{1}{2}$ ) prior may be more appropriate over complete ignorance imposed by a Beta(1, 1) prior. The former distribution is U-shaped symmetric and places greater probability density at the extremes of the distribution due to its heavier tails, while still allowing for variability in parameter estimates within intermediate values along its domain. Note that this prior is Jeffreys' prior density (Jeffreys, 1946), which is proportional to the square root of the

661 Fisher information  $\mathcal{I}(\theta)$ ; that is,  $\pi(\theta) \propto \theta^{-\frac{1}{2}}(1 - \theta)^{-\frac{1}{2}}$ . Jeffreys' prior has several desirable  
 662 statistical properties as a prior: that it is inversely proportional to the standard deviation of  
 663 the binomial distribution, and most notably, that it is invariant to model reparameterization  
 664 (Gelman et al., 2014). However, this prior can lead to divergent transitions, among other  
 665 pathologies, imposed by complex geometry (*i.e.*, curvature) in the posterior space since many  
 666 iterative stochastic MCMC sampling algorithms experience difficulties when exploring high  
 667 density distribution regions. Thus, remedies to resolve them, such as lowering the step size of  
 668 the HMC/NUTS sampler, should be attempted in future work, along with other approaches  
 669 such as empirical Bayes estimation to approximate beta prior hyperparameters from observed  
 670 data through the MLE or other methods of parameter estimation, such as the method  
 671 of moments. Alternatively, hierarchical modelling could be employed to estimate separate  
 672 distribution model hyperparameters for each species and/or compute distinct estimates for  
 673 the directionality/comparison level of the DNA barcode gap metrics (*i.e.*, lower *vs.* upper,  
 674 non-prime *vs.* prime) separately within the genus under study. This would permit greater  
 675 flexibility through incorporating more fine-grained structure seen in the data; however, low  
 676 taxon sample sizes may preclude valid inferences to be reasonably ascertained due to the  
 677 large number additional parameters which would be introduced through the specification  
 678 of the hyperprior distributions. Methods outlined in Gelman et al. (2014, 2020), such as  
 679 dealing with non-exchangeability of observations and alternate model parameterizations like  
 680 the logit, may prove useful in this regard. Perhaps, the best course of action though is  
 681 the elicitation of priors from domain experts. Even though more work remains, it is clear  
 682 that both frequentist and Bayesian inference hold much promise for the future of molecular  
 683 biodiversity science.



## Supplementary Information

None declared

## Data Availability Statement

Raw data, R, and Stan code can be accessed via Dryad at:

<http://datadryad.org/stash/share/>

RZIfMixcEODe0RWP7eyXWQewSVbqEIA9UTrH3ZVKyn4.

A GitHub repository can be found at:

<https://github.com/jphill01/Bayesian-DNA-Barcode-Gap-Coalescent>.

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## Conflict of Interest

None declared.

## Author Contributions

JDP wrote the manuscript, wrote R and Stan code, as well as analyzed and interpreted all model results.

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